

QA API Test Automation Assignment

Objective

Build a production ready API automation framework using your preferred stack. Your test suite will validate stateful, dependent, and long-running asynchronous endpoints on a local mock server.

Setup Instructions

1. Download or clone the mock server script from the Gist:
<https://gist.github.com/sharanya-lb/b429b6e807f95a8df8216c4343ff6766>
2. Run the server locally:

```
npm install express  
node server.js
```
3. Server Base URL:
`http://localhost:3000/v1`

Target Endpoints & Business Logic

Endpoint	Method	Input Parameters	Key Behaviors to Validate
/v1/auth/login	POST	Body: { username, apiKey }	Returns Bearer JWT token required for protected routes.
/v1/orders	POST	Header: X-Correlation-ID Body: { customerId, items, shippingAddress }	Creates order (202 Accepted). Dynamically calculates totalAmount = sum(quantity * unitPrice). Initial status: PENDING.
/v1/orders/:orderId	GET	Bearer Token	State transition engine: Status auto-shifts PENDING →

Endpoint	Method	Input Parameters	Key Behaviors to Validate
			PROCESSING (5s) → COMPLETED (15s).
/v1/orders/:orderId	DELETE	Bearer Token	Cancels order. Allowed during PENDING or PROCESSING. Returns 409 Conflict if state is COMPLETED.
/v1/exports	POST	Bearer Token	Triggers long-running CSV generation. Returns 202 Accepted and jobId.
/v1/exports/:jobId	GET	Bearer Token	Status stays PROCESSING for 60 seconds before transitioning to COMPLETED.
/v1/exports/:jobId/download	GET	Bearer Token	Returns 200 OK, Content-Type: text/csv, and raw CSV payload. Returns 400 if the job is incomplete.

Technical Requirements & Submission

- **Endpoint Coverage:** Your test suite must fully cover **all endpoints** listed above, validating expected behaviors, HTTP status codes etc.,
- **Submission Format:** Submit your solution as a public GitHub repository link containing your fully working automation framework. Include clear instructions in your repository's README.md detailing how to execute the test suite.